SIX EQUIVALENT FORMULATIONS OF AMPLENESS OVER AN AFFINE BASE IN FINITE TYPE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. Read in full under the Lead's 12:4x rule. The proof contains 'Trivially (6) implies (5)' / 'Trivially (4) implies (3)' - the word marks genuinely trivial implications between the listed conditions, and every non-trivial step is a cited result, so the assembly of the equivalence is the author's own work.

Allowed standard prerequisites (editorial).

- Ampleness on an affine total space equals relative ampleness. Lemma \ref{lemma-ample-over-affine}, cited at proof line 9538. Identifies the first two conditions.
- Relative ampleness gives an immersion into projective space for finite type morphisms over affine bases. Lemma \ref{lemma-quasi-projective-finite-type-over-S}, cited at proof lines 9539-9540. Gives condition (6).
- Very ampleness implies ampleness, and ampleness is stable under taking powers. Lemma \ref{lemma-ample-very-ample} and Lemma \ref{lemma-ample-power-ample}, cited at proof lines 9548-9549. Close the cycle.

Source reference. Original lemma 01VT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2458–2458. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01VK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2449–2449. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01VS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2457–2457. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01OE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2191–2191. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01VN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2452–2452. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02NN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3486–3486. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module. Assume S affine and f of finite type. The following are equivalent*

(1) *$\mathcal{L}$ is ample on X,*
(2) *$\mathcal{L}$ is f-ample,*
(3) *$\mathcal{L}^{\otimes d}$ is f-very ample for some $d \geq 1$,*
(4) *$\mathcal{L}^{\otimes d}$ is f-very ample for all $d \gg 1$,*
(5) *for some $d \geq 1$ there exist $n \geq 1$ and an immersion $i : X \to \mathbf{P}^n_S$ such that $\mathcal{L}^{\otimes d} \cong i^*\mathcal{O}_{\mathbf{P}^n_S}(1)$, and*
(6) *for all $d \gg 1$ there exist $n \geq 1$ and an immersion $i : X \to \mathbf{P}^n_S$ such that $\mathcal{L}^{\otimes d} \cong i^*\mathcal{O}_{\mathbf{P}^n_S}(1)$.*

Proof. The equivalence of (1) and (2) is Lemma 01VK. The implication (2) $\Rightarrow$ (6) is Lemma 01VS. Trivially (6) implies (5). As $\mathbf{P}^n_S$ is a projective bundle over S (see Constructions, Lemma 01OE) we see that (5) implies (3) and (6) implies (4) from the definition of a relatively very ample sheaf. Trivially (4) implies (3). To finish we have to show that (3) implies (2) which follows from Lemma 01VN and Lemma 02NN. □